

## Project Title:

### Image Processing Pipeline using AWS SQS

## Project Description:

Designed and implemented a **decoupled image processing pipeline** using **Amazon Simple Queue Service (SQS)** to handle asynchronous tasks.

The system processes uploaded images by resizing them using a worker running on **Amazon EC2**, with images stored in **Amazon S3**.

## Architecture Overview

User Upload → S3 (Input Bucket)



SQS Queue



EC2 Worker (Python)



Processed Image → S3 (Output Bucket)

## AWS Services Used

- Amazon S3 → Object storage for images
- Amazon Simple Queue Service → Message queue for task processing
- Amazon EC2 → Worker instance for processing
- AWS Identity and Access Management → Role-based access

## Implementation Steps

### 1. Created S3 Buckets

- Input bucket for uploading images
- Output bucket for storing processed images

### 2. Configured SQS Queue

- Created a Standard Queue
- Used to send image processing tasks

### 3. IAM Role Setup

- Created IAM role for EC2
- Attached policies:
  - AmazonS3FullAccess
  - AmazonSQSFullAccess

#### 4 Launched EC2 Instance

- Ubuntu-based EC2 instance
- Attached IAM role
- Installed Python, boto3, Pillow

#### 5 Developed Worker Script

Key functionalities:

- Poll messages from SQS
- Extract image name from message
- Download image from S3
- Resize image using Pillow
- Upload processed image to S3
- Delete message from queue

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#### Sample Logic

```
image_key = message['Body'].strip()
```

```
s3.download_file('input-bucket', image_key, image_key)
```

```
img = Image.open(image_key)
```

```
img = img.resize((200, 200))
```

```
processed = "processed-" + image_key
```

```
img.save(processed)
```

```
s3.upload_file(processed, 'output-bucket', processed)
```

## Challenges Faced & Fixes

### ❌ S3 404 Error

- Cause: Incorrect object key / file not uploaded
- Fix: Verified exact file name and uploaded properly

### ❌ AWS CLI Missing

- Cause: EC2 did not have AWS CLI installed
- Fix: Installed AWS CLI manually

### ❌ Message Formatting Issue

- Cause: Extra spaces/newline in SQS message
- Fix: Used `.strip()`

### ❌ Bucket Name Mismatch

- Cause: Incorrect bucket name in script
- Fix: Updated with correct bucket

## Key Features

Asynchronous processing using SQS

Decoupled architecture

Scalable worker-based design

Automated image processing pipeline

## Outcome

- Successfully processed images using queue-based system
- Reduced dependency between upload and processing components
- Built a real-world DevOps-style pipeline

## What I Learned:

Working with AWS services integration

Debugging cloud-based pipelines

Handling asynchronous workflows

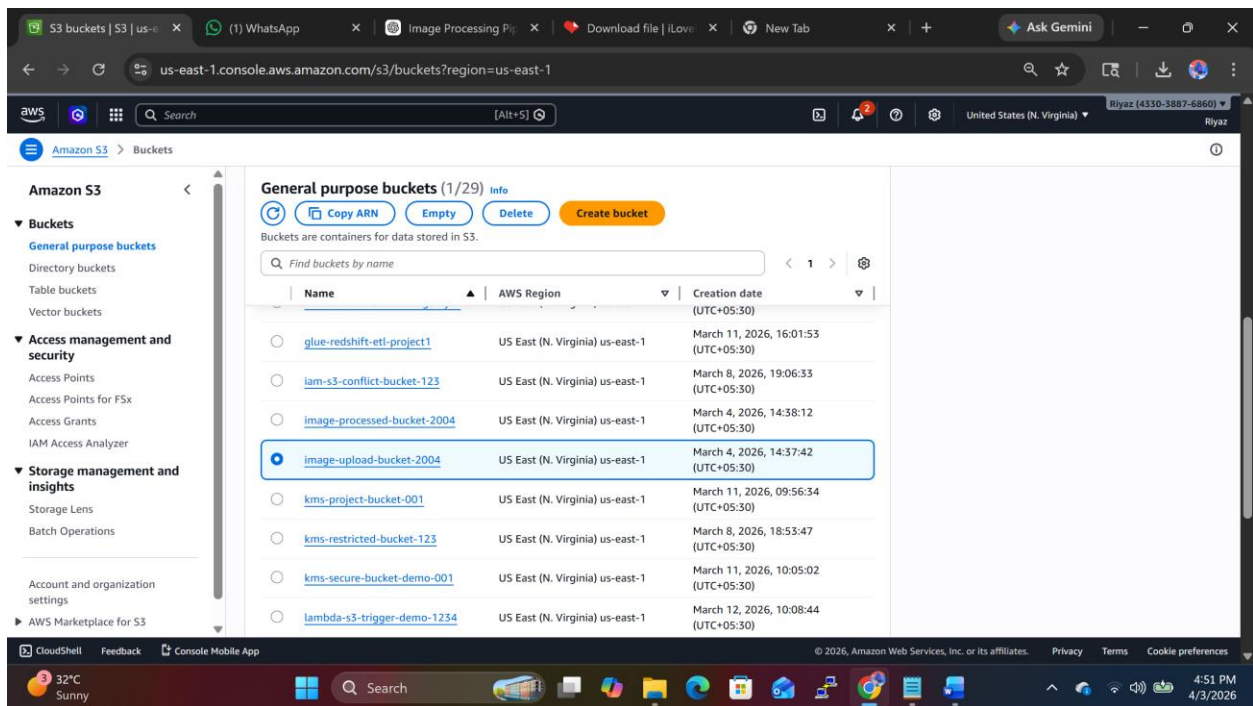
IAM role-based permissions

Real-world DevOps troubleshooting

S3 buckets created:

Image-upload-bucket-2004

Image-processed-bucket-2004



Inside buckets:

The screenshot shows the AWS S3 console interface for the bucket 'image-upload-bucket-2004'. The left sidebar contains navigation links for Buckets, Access management and security, and Storage management and insights. The main content area displays the 'Objects' tab with a table listing the objects in the bucket. There is one object, 'parrot.jpg', which is a 7.2 KB file of type 'jpg' uploaded on March 4, 2026, at 15:27:50 (UTC+05:30). The storage class is 'Standard'.

| Name       | Type | Last modified                       | Size   | Storage class |
|------------|------|-------------------------------------|--------|---------------|
| parrot.jpg | jpg  | March 4, 2026, 15:27:50 (UTC+05:30) | 7.2 KB | Standard      |

The screenshot shows the AWS S3 console interface for the bucket 'image-processed-bucket-2004'. The left sidebar contains navigation links for Buckets, Access management and security, and Storage management and insights. The main content area displays the 'Objects' tab with a table listing the objects in the bucket. There is one object, 'processed-parrot.jpg', which is a 7.0 KB file of type 'jpg' uploaded on March 4, 2026, at 15:29:08 (UTC+05:30). The storage class is 'Standard'.

| Name                 | Type | Last modified                       | Size   | Storage class |
|----------------------|------|-------------------------------------|--------|---------------|
| processed-parrot.jpg | jpg  | March 4, 2026, 15:29:08 (UTC+05:30) | 7.0 KB | Standard      |

Sqs created:

us-east-1.console.aws.amazon.com/sqs/v3/home?region=us-east-1#/queues

Amazon SQS > Queues

Queues (15)

Search queues by prefix

Buttons: Edit, Delete, Send and receive messages, Actions, Create queue

| Name                                     | Type     | Created                | Messages available | Messages in flight | Encryption               | Content-based deduplication |
|------------------------------------------|----------|------------------------|--------------------|--------------------|--------------------------|-----------------------------|
| <a href="#">analytics-queue</a>          | Standard | 2026-03-04T17:27+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">dlq-queue</a>                | Standard | 2026-03-05T09:13+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">email-notification-queue</a> | Standard | 2026-03-04T12:28+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">email-queue</a>              | Standard | 2026-03-04T17:27+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">image-processing-queue</a>   | Standard | 2026-03-04T14:39+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">lambda-dlq-queue</a>         | Standard | 2026-03-12T16:55+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">lambda-sqs-queue</a>         | Standard | 2026-03-12T09:27+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">large-message-queue</a>      | Standard | 2026-03-05T12:30+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">main-processing-queue</a>    | Standard | 2026-03-05T09:16+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">microservices-queue</a>      | Standard | 2026-03-04T15:41+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">notification-queue</a>       | Standard | 2026-03-04T18:59+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |
| <a href="#">order-processing-queue</a>   | Standard | 2026-03-04T14:09+05:30 | 0                  | 0                  | Amazon SQS key (SSE-SQS) | -                           |

CloudShell Feedback Console Mobile App

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Final output:

```
ubuntu@ip-172-31-11-105: ~  
ubuntu@ip-172-31-11-105:~$ python3 worker.py  
Received image key: parrot.jpg  
Image processed: processed-parrot.jpg
```

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